Geometry Honors Unit 6 Part 2 Assessment Guide

General Information	This assessment covers the benchmarks from lessons 7-12. Although MA.912.GR.1.2 and MA.912.GR.1.6 are not related, both are connected to the benchmarks in lessons 1-6. The pieces of these benchmarks related to congruent figures were covered in Unit 5. Students extend their understanding of these benchmarks to the similar figures in Unit 6. In grade 7, students solved mathematical and real-world problems involving scale drawings and scale factors (MA.7.GR.1.5). Students who struggle with solving problems using similarity may find reviewing Grade 7, Unit 5, lessons 8-10 helpful. Teachers may also want to consider that students who struggle with solving problems using similarity may struggle with proportions. Students who show an inability to set up proportions and/or solve proportions may benefit from lessons from Grade 7, Units 6 or 11. Students will continue to utilize their understanding of similarity throughout the remaining units. Teachers should consider how to support any misconceptions students may have so that the student can correctly apply the concepts of similarity in related learning. This honors assessment also assesses benchmark MA.912.GR.2.9, where students use their understanding of similarity to justify similarity in terms of non-rigid motions.
Calculator Information	Students should be given the use of a calculator on this assessment.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.GR.1.2
AA similarity		**Recommended Scoring** **(6 Total Points)**	No partial credit is suggested.
Question	**2a**	**Benchmark(s)**	MA.912.GR.1.2
a. $\frac{BT}{BD} = \frac{BW}{BR}$ | ΔDBR and ΔTBW | ○ Yes ○ No | SAS similarity		**Recommended Scoring** **(7 Total Points)**	Consider 3 points for yes and 4 points for SAS similarity. Give 3 points of the 4 if student only states SAS.
Question	**2b**	**Benchmark(s)**	MA.912.GR.1.2
b. $\overleftrightarrow{BX} \parallel \overleftrightarrow{WL} \parallel \overleftrightarrow{MD}$ and $\frac{BW}{WR}$ | ΔTWB and ΔTWR | ○ Yes ○ No |		**Recommended Scoring** **(7 Total Points)**	No partial credit suggested.
Question	**2c**	**Benchmark(s)**	MA.912.GR.1.2
c. $\overleftrightarrow{BX} \parallel \overleftrightarrow{MD}$ | ΔYTB and ΔRTD | ○ Yes ○ No | AA similarity		**Recommended Scoring** **(7 Total Points)**	Consider 3 points for yes and 4 points for AA similarity. Give 3 points of the 4 if student only states AA.

Question	2c	Benchmark(s)	MA.912.GR.1.2
d. $\frac{BR}{WR}=\frac{YR}{TR}=\frac{YB}{TW}$ | ΔYRB and ΔTRW | ○ Yes ○ No | SSS similarity		Recommended Scoring (7 Total Points)	Consider 3 points for yes and 4 points for SSS similarity. Give 3 points of the 4 if student only states SSS.
Question	**3a**	**Benchmark(s)**	MA.912.GR.2.8 MA.912.GR.2.9
(see table below)		Recommended Scoring (8 Total Points)	Consider 4 points for each algebraic description.

Sequence	Algebraic Description
Transformation 1	$(x, y) \rightarrow (x, -y)$
Transformation 2	$(x, y) \rightarrow (3x, 3y)$

Question	3b	Benchmark(s)	MA.912.GR.2.9
The sequence of transformations that map ΔHRB onto $\Delta H''R''B''$ include a reflection and a dilation. Because the sequence is a rigid motion and a dilation, the definition of similarity in terms of non-rigid transformations states that the triangles are similar.		Recommended Scoring (4 Total Points)	Consider 4 points for a correct justification using the definition of similarity in terms of non-rigid transformations.

Question	4a-c	Benchmark(s)	MA.912.GR.1.6
<table><tr><th></th><th>Point</th><th>work</th><th>Height (cm)</th></tr><tr><td>a.</td><td>M</td><td>$\frac{6.3}{10.5}=\frac{x}{70}$</td><td>42</td></tr><tr><td>b.</td><td>G</td><td>$\frac{6.3}{10.5}=\frac{x}{122.5}$</td><td>73.5</td></tr><tr><td>c.</td><td>X</td><td>$\frac{6.3}{10.5}=\frac{x}{196}$</td><td>117.6</td></tr></table>		**Recommended Scoring** **(18 Total Points)**	Consider 6 points for each part - 3 points for any correct set up of the proportion and 3 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**5a**	**Benchmark(s)**	MA.912.GR.1.6
$\frac{85}{BR}=0.32$ $BR=265.625$		**Recommended Scoring** **(6 Total Points)**	Consider 3 points for any correct set up and 3 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.

Question	5b	Benchmark(s)	MA.912.GR.1.6
$\frac{(11x-19.5)}{(29.5x-19.5)} = 0.32$ $\quad x = 8.5$		**Recommended Scoring** **(12 Total Points)**	Consider 6 points for any correct set up and 6 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**5c**	**Benchmark(s)**	MA.912.GR.1.6
$HY = 6(8.5) + 4 = 55$ $\frac{HY}{HB} = 0.32 \quad \frac{55}{HB} = 0.32 \quad HB = 171.875$ $BY = 171.875 - 55 = 116.875$		**Recommended Scoring** **(18 Total Points)**	Consider distributing points as follows: • 6 points for HY, 3 points for correct set up and 3 points for finding the value • 6 points for HB, 3 points for correct set up and 3 points for finding the value • 6 points for BY, 3 points for correct set up and 3 points for finding the value Consider student work when determining partial credit. If a student incorrectly finds the value of x in part b, they should not be penalized twice if they use that value correctly in their work for part c.